Low Voltage Bandgap Reference using Intelligent Layout Generators in 22 nm FDSOI CMOS

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***Abstract*—This paper presents a low-voltage bandgap reference (BGR) circuit fabricated using 22 nm CMOS fully-depleted silicon on insulator (FDSOI) technology, incorporating an adaptive body-basing technique. During the design phase, the g_m/I_D methodology and the Intelligent IP (IIP) generators are utilized as essential reference strategies. The circuit occupies a chip area of 0.044 mm^2 and consumes only 1.8 µA of current from a 1.5 V DC supply voltage. It provides an output reference voltage of 610 mV, with a temperature coefficient (TC) of 37 ppm/°C. Additionally, it provides a current reference of 0.4 µA, with a temperature coefficient of 60 ppm/°C, and operates effectively over a wide temperature range of −40 °C to 125 °C. Furthermore, the proposed circuit offers a power supply rejection ratio of 63 dB, making it a suitable solution for low-power, stable reference voltage applications.**

***Index Terms*—voltage reference, low-voltage cascode, layout automation, temperature coefficient, power supply rejection ratio.**

I. Introduction

Interest in low power, high precision bandgap voltage, and current references has increased recently, especially in battery-powered medical, pharmaceutical, and environmental monitoring devices [1], [2]. Fig. 1 shows a simplified block diagram of the bandgap reference (BGR) application in integrated circuit (IC) systems as part of the power management unit (PMU). It provides a reference voltage for low drop-out voltage regulators (LDOs) [3], including voltage/current reference levels for subsequent microwave sensors [4], digital controllers [5], analog to digital converters (ADC) [6], RF receivers (RX), and transmitter (TX) blocks making the bandgap circuit a necessary building block in any IC systems.

Modern down-scaled technologies have scaled power supplies to sub 1-V to meet transistor reliability and ensure low power consumption. A low-voltage realization of bandgap circuits was first proposed in [7], where the authors combine a Proportional to Absolute Temperature (PTAT) and a Complementary to Absolute Temperature (CTAT) currents to get a zero-temperature voltage reference across a resistor. Numerous variations and improvements of PTAT and CTAT current combination techniques have been proposed in recent years [8]–[10]. In work [8], authors investigate the influence of process, voltage, and temperature (PVT) mismatch caused by BJT and resistor arrays on the output reference voltage. They propose a low-voltage BJT-based bandgap circuit with an automatic feedback loop to improve reference voltage accuracy.

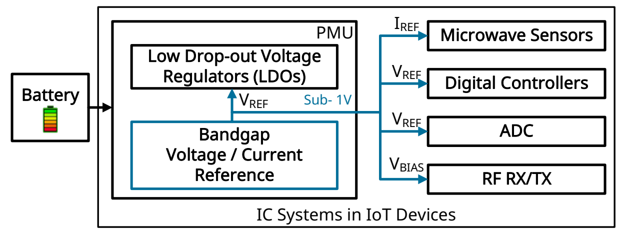

Fig. 1. Simplified block diagram of the bandgap application in IC systems.

The work [9] proposes a current-based bandgap circuit with an auto-calibration technique using a current digital-to-analog converter as a part of a control loop. However, using such calibration and trimming circuits increases the complexity and leads to chip area costs. A compact-size current-based bandgap circuit was proposed in [10], where a zero-temperature coefficient MOSFET transistor was used as curvature compensation. At the same time, the power consumption was achieved at 77 µW from a single supply.

In this paper, we present a low-power BJT-based bandgap voltage and current reference circuit using the $\mathrm{g_m/I_D}$ methodology to accurately describe the behavior of the MOSFET transistors in the weak and moderate inversion regions. Furthermore, the design process benefits from the Intelligent IP (IIP) generator approach [11], which enables rapid exploration of layout configurations and optimal device partitioning, reducing design effort and enhancing overall performance. The circuit is realized as a cascode structure with an adaptive body biasing technique enhancing the power supply rejection ratio (PSRR), and it has been implemented using 22 nm FDSOI CMOS technology.

II. Circuit Design

Optimization of transistors is essential in low-power operations, where leveraging the $\mathrm{g_m/I_D}$ ratio enables precise and efficient characterization in the weak and moderate inversion regions. The relationship is defined as the derivative of the logarithm of $\mathrm{I_D}$ with respect to $\mathrm{V_G}$ as shown in [12]:

$$\frac{\mathrm{g_m}}{\mathrm{I_D}} = \frac{1}{\mathrm{I_D}}\frac{\partial \mathrm{I_D}}{\partial \mathrm{V_G}} = \frac{\partial \log(\mathrm{I_D})}{\partial \mathrm{V_G}} \tag{1}$$

The dependence of $\mathrm{I_D}$ on the overdrive voltage ($\mathrm{V_{OV}}$) is simulated for both NMOS and PMOS transistors with a size of $W/L = 2\,\mu m/2\,\mu m$ as illustrated in Fig. 2. Both transistors achieve a maximum g_m/I_D beyond $30\,mS/A$ when V_{OV} is in the range of −0.3 to −0.1 V. This corresponds to the weak inversion region, which offers the highest current efficiency as implied by (1). The derivative reaches its peak in this operation region, where the drain current falls into nA range which is in favour of low-power designs.

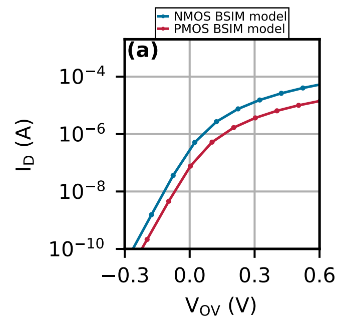

Fig. 2. (a) I_D and (b) g_m/I_D characteristic simulations using BSIM model for the given PDK in 22FDSOI process. Transistors sizes are 2 µm/2 µm.

A. Proposed Bandgap Circuit

Fig. 3 illustrates the proposed bandgap circuit. The circuit consists of a BGR core ($Q_{1,2}$, R_{1-3}, M_{1-8}), a start-up circuit (M_{S1-S3}, R_{S1}, Q_{S1}), a biasing circuit (M_{B1-B6}, R_{B1-B3}, $C_{B1,B2}$, Q_{B1}), a current reference ($M_{11,12}$) and a voltage reference ($M_{9,10}$, R_4) with a low-pass filter (R_5, M_N). The design parameters of both active and passive devices are carefully chosen based on the aforementioned g_m/I_D approach and are summarized in Table I. In low-voltage reference circuit designs, bipolar transistors $Q_{1,2}$ provide a more reliable and predictable V_{BE} compared to the significantly varying threshold voltage of MOSFETs across PVT. The BGR reference voltage is given as:

$$V_{REF} = R_4 \left(\frac{\Delta V_{BE}}{R_2} + \frac{V_{BE1}}{R_3} \right) < 1.2\,V \qquad (2)$$

To achieve low PVT sensitivity while minimizing power consumption, the BGR core employs BJTs and low-voltage cascode current mirrors. This design eliminates the need for operational amplifiers, reducing power consumption and

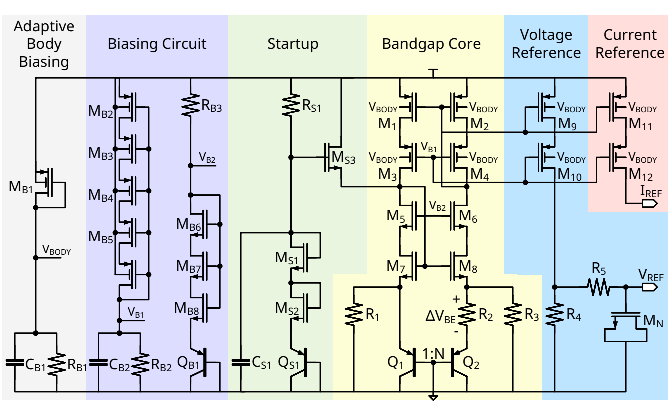

Fig. 3. Proposed bandgap reference circuit.

maintaining low line sensitivity. Additionally, the cascode structure enhances the PSRR of the core by increasing the output impedance.

TABLE I
LISTING THE DEVICE SIZING AND VALUES

Component	W / L	Component	Value
M_{B1}	80×4 µm / 0.1 µm	R_{B1}	15.4 MΩ
M_{B2-B5}	2 µm / 0.2 µm	R_{B2}	3.33 MΩ
M_{B6-B8}	2 µm / 2 µm	R_{B3}	5.4 MΩ
Q_{B1}	2 µm / 2 µm	R_{S1}	5.4 MΩ
M_{S1-S2}	8 µm / 0.8 µm	$R_{1,3}$	3.08 MΩ
M_{S3}	2×4 µm / 1 µm	R_2	388 kΩ
Q_{S1}	4 µm / 4 µm	R_4	1.5 MΩ
$M_{1,2,9,11}$	64×4 µm / 2 µm	R_5	100 kΩ
$M_{3,4,10,12}$	64×4 µm / 1 µm	$C_{B1,B2}$	1.1 pF
$M_{5,6}$	64×2 µm / 1 µm	C_{S1}	0.5 pF
$M_{7,8}$	64×2 µm / 2 µm		
Q_1	3 µm / 3 µm		
Q_2	16×3 µm / 3 µm		
M_N	16×8 µm / 8 µm		

Notably, the body of M_{1-4} and M_{9-12} are biased with adaptive body-biasing technique designed to dynamically compensate the threshold voltage of the transistors, enabling reliable operation in the aforementioned weak inversion region across temperature variations. As V_{BODY} increases with temperature, the threshold voltage of the body-biased MOSFETs decreases, preserving their operation in the weak inversion region and thus enhancing current efficiency. This technique further reduces the temperature coefficient of the reference voltage and current while improving the PSRR. A zero-TC current is then generated from the BGR core through carefully designed R_{1-3}, which is then mirrored and passed through R_4 to generate the reference voltage, the reference current is accurately replicated by $M_{4,8}$ to ensure stable operation.

Post-layout Monte-Carlo analysis is conducted to investigate the impact of device mismatch on the distribution of the reference voltage and current, as shown in Fig. 4. A total of 1000 samples are simulated, yielding 1.28 % for V_{REF} at 600 mV and 5.15 % for I_{REF} at 400 nA.

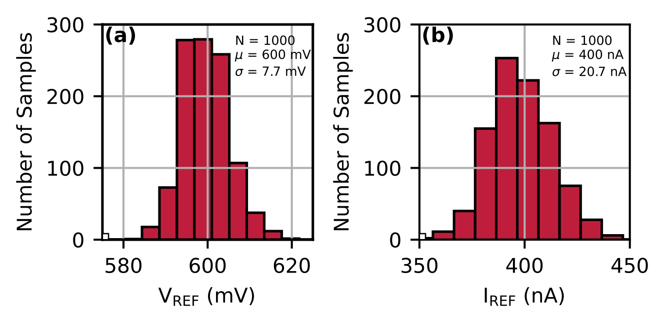

Fig. 4. (a) V_{REF} and (b) I_{REF} distribution of the post-layout bandgap circuit at room temperature.

B. Layout Automation with the IIP Tool Procedure

Due to the advanced requirements for accuracy in many parts of the circuit it can be assumed that besides PVT also a proper selection of possible layout implementation variants

TABLE II
$\mathrm{M}_{1,2,9,11}$ TRANSISTOR GROUP PLACEMENT USING THE IIP TOOL

Placement style	N-rows	Area	$\mathrm{V}_{\mathrm{REF},\mu}$	μ/σ
Side-by-side	4	280×11 µm	585 mV	1.45 %
Side-by-side	8	150×25 µm	605 mV	1.39 %
Interdigitated	8	150×25 µm	600 mV	1.28 %

will have a significant influence on the overall performance. Usually, the number of investigated layout variants is limited by the effort creating them, unless there is an automated solution. For this work, we applied the Intelligent IP generator approach [11] for the implementation of the low voltage cascode current mirrors as they consume a significant part of the chip area and are important for the accurate operation of the bandgap core and the reference output stages.

IIP can create schematic, symbol, and layout cellviews for several types of basic building blocks. The results can be controlled by parameters for topology variants, device types, sizing, and several layout options for placement and routing. It would also ease the migration of the design between PDKs of different nodes or vendors due to a separate mapping of its generic core to process-specific properties such as devices, layers, and design rules. This way, the same IIP generator instances can simply be re-executed in another PDK to get corresponding and rule-compliant design data. For more complex designs, IIP can even generate code based on existing schematics for creating new generators including a template-based layout engine [13].

Here, we used the basic MosArray generator of the IIP library already during schematic phase by generating the mirror and cascode devices of the bandgap core together with the MOS devices of the voltage and current references into dedicated subblocks. As the corresponding layout cellviews were generated at the same time within seconds, this made sizing decisions visible immediately and already helped e.g. with selecting a proper partitioning of large device widths into fingers and multipliers. For fine-tuning of the layout, we investigated three layout variants of the mirror MOS transistor array with its 256 elements where we used different place patterns and numbers of rows, resulting in different aspect ratio, area and amount of routing for the resulting layouts. Fig. 5 shows placement styles of $\mathrm{M}_{1,2,9,11}$ group of transistors. The post-layout Monte-Carlo simulation results are shown in Table II and allowed us to identify the best of these variants for final implementation. Without the layout generation tool we would not have implemented several variants and thus might have chosen a sub-optimal one. Also, we saved an estimated effort of 3 days for the conventional layout implementation.

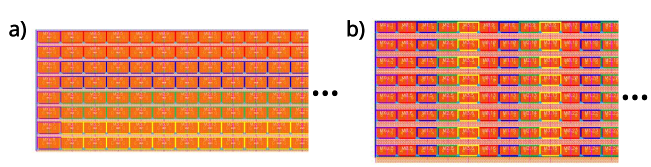

Fig. 5. (a) side-by-side and (b) interdigitated placement styles of $\mathrm{M}_{1,2,9,11}$ using the MosArray IIP generator.

Further MosArray layout options and generators for other common structures including passives are currently developed or already available and will be used in a next design.

III. Measurement Results

The proposed bandgap reference circuit has been designed and manufactured in CMOS 22 nm FDSOI technology from Globalfoundries. The fabricated chip photo and its corresponding layout are shown in Fig. 6. The active area of the proposed bandgap circuit occupies 0.044 mm^2. The BJT core is realized using a common centroid layout to minimize the mismatch effects and PVT variations. Due to the low-power implementation, poly-resistors and MOS transistors occupy a significant portion of the layout area.

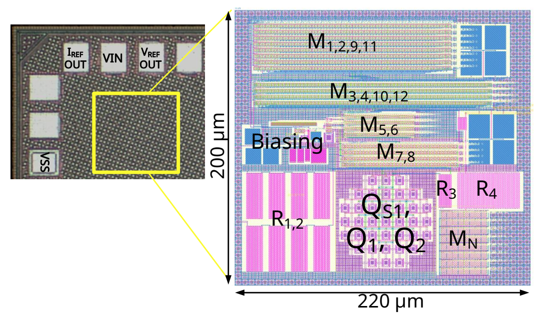

Fig. 6. Chip photo and layout of the BGR with an active area of 0.044 mm^2.

The chip was measured in a SUSS MicroTec probe station with a chuck heating/cooling temperature control with a C40 ATT system to sweep the temperature from −40 °C to 125 °C. The DC measurement of the proposed bandgap circuit was performed using an HP4142B SMU Modulator DC supply. As illustrated in Fig. 7, the reference voltage and current are measured across temperature and maintaining steady output levels over an input voltage range of 1.3 to 1.8 V.

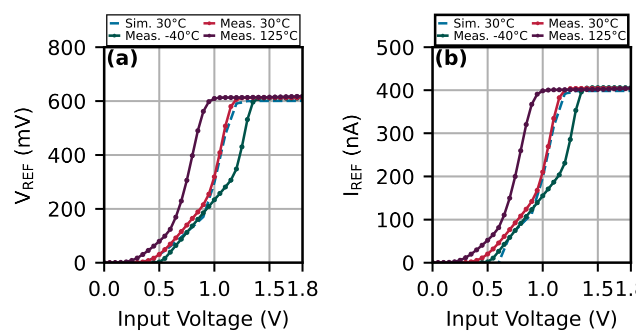

Fig. 7. Measured reference (a) voltage and (b) current of the BGR over input voltage.

Fig. 8a and Fig. 9a present the measured output voltage and current references at different input supply voltages. A stable reference of 610 mV and 405 nA is observed across temperature. As shown in Fig. 8b and Fig. 9b, the TC of the voltage reference and the current reference reaches 37 ppm/°C and 60 ppm/°C at the nominal input voltage, respectively.

The PSRR of the BGR measurement and simulation results is presented in Fig. 10. The measured result indicates that the

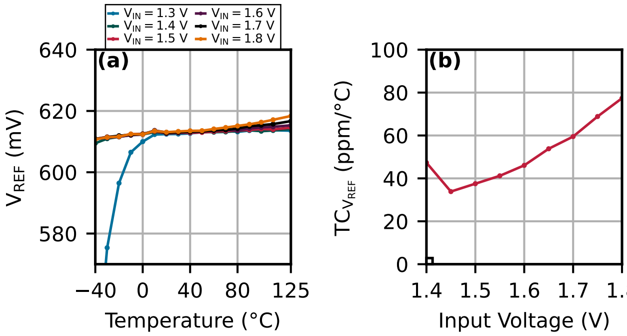

Fig. 8. Measured (a) reference voltage over temperature and (b) its temperature coefficient $TC_{V_{REF}}$ over input voltage.

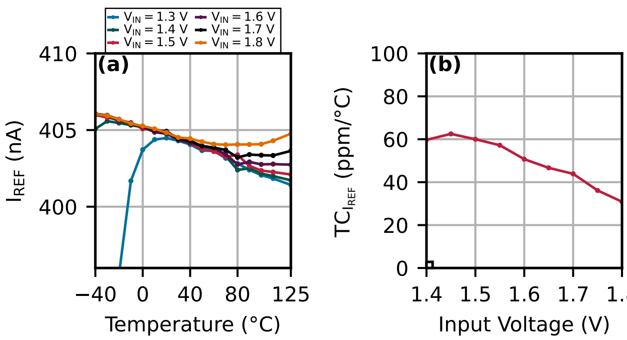

Fig. 9. Measured (a) reference current over temperature and (b) its temperature coefficient $TC_{I_{REF}}$ over input voltage.

proposed design achieves an input noise suppression of 63 dB at 10 Hz. The performance of the proposed bandgap voltage reference circuit is summarized in Table III and compared with other state-of-the-art bandgap designs.

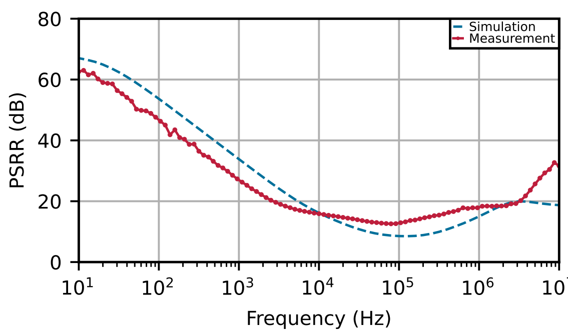

Fig. 10. Measured power supply rejection ratio (PSRR) at V_{IN} = 1.5 V.

IV. Conclusion

In this paper, a low-power low-voltage BJT-based bandgap reference circuit implemented in 22 nm FDSOI technology is presented. The circuit employs the adaptive body-biasing technique to maintain stable voltage and current references while further reducing the TC. A cascode structure enhances supply rejection, and the g_m/I_D methodology optimizes power consumption and chip area. Moreover, the IIP generator streamlined the layout generation process, further improving design efficiency. The overall performance of the proposed design makes it a competitive solution for low-power PMUs compared to the state-of-the-art.

TABLE III
Performance summary and comparison with state of the art.

	This work	[8] ISSCC	[9] A-SSCC	[10] TCAS-II
Technology	**22 nm**	12 nm	65 nm	180 nm
Type	**BJT**	BJT	BJT	BJT
V_{IN}, V	**1.3–1.8**	0.7–1.5	1.2–2.5	1.2–2.4
V_{REF}, V	**0.61**	0.21	1	0.628
I_{REF}, A	**0.405 µ**	-	-	-
I_{DC}, A	**1.8 µ**	9.8 µ	0.81 µ	64 µ
T_{Range}, °C	**–40–125**	–20–125	–40–120	–40–120
σ/μ, %	**1.28**[a]	2.35	0.35	0.33
$TC_{V_{REF}}$ (ppm/°C)	**37**	40.5	28.8	2.5
$TC_{I_{REF}}$ (ppm/°C)	**60**	-	-	-
PSRR (dB)	**63 @10 Hz**	-	58 @DC	91[a] @100 Hz
Trimming	**No**	No	Yes	Yes
Active area (mm^2)	**0.044**	0.065	0.04	0.024

a: Post-layout simulation.

Acknowledgment

This work was supported in part by UNLOOC HORIZON-KDT-JU-2023-IA-101140192 and in part by the German BMBF funded project HoLoDEC 16ME0704.

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